

NKY 225 Enhanced Index

ML Alpha Generation & Portfolio Optimisation

Market-neutral active weight strategy · Data window: 2014-01-01 → present
Benchmark weights & constituent membership sourced from JPY121 ETF (1321.T)

STEP 1 DATA COLLECTION

- ① NKY 225 Index Price — Yahoo Finance (^N225), J-Quants, Stooq | From: 2014-01-01
- ② JPY121 ETF (1321.T) Holdings — daily constituent weights & in/out membership events
- ③ Constituent OHLCV — yfinance (.T suffix), J-Quants, Alpha Vantage | ALL ever-members
- ④ Fundamentals — EDINET (FSA), J-Quants /fins/statements, SimFin
- ⑤ Macro & Regime — FRED (fredapi), BOJ API, OECD, Cabinet Office Japan
- ⑥ Sentiment — Nikkei VI (JPX), CBOE VIX via FRED, TSE short interest
- ⑦ Calendar & Cross-Section — derived from price data & dates



STEP 2 RAW FACTOR STORE

- ~300 raw factors across: Momentum (41) · Volatility (37) · Volume/Liquidity (25)
- Fundamental (44) · Macro & Regime (44) · Sentiment (19) · Cross-Section (14) · Calendar (21)



STEP 3 SURVIVORSHIP-BIAS-FREE UNIVERSE & BENCHMARK WEIGHTS (JPY121 ETF)

- ① Download 1321.T (JPY121 ETF) daily holdings from 2014-01-01
- ② Build constituent panel: date × stock → {in_index, w_bench_i (price-weighted)}
- ③ Flag ADDED events (entry date) and REMOVED events (exit date + delisting return)
- ④ Universe = all stocks ever in ETF holdings → ~270-300 unique stocks (bias-free)

STEP 4 POINT-IN-TIME (PIT) ALIGNMENT

- ① Fundamental data stamped at announcement date (not period end) — prevents look-ahead
- ② Use J-Quants announcement_date field or EDINET disclosure timestamps
- ③ Stocks removed from index: features retained up to removal date only

**STEP 5 OUTLIER WINSORISATION**

- ① Winsorise each raw factor at 1st and 99th percentile — cross-sectional, per date
- ② Prevents extreme single observations from dominating the normalised panel

**STEP 6 MISSING VALUE IMPUTATION**

- ① Price data: forward-fill up to 5 trading days (covers halted / suspended stocks)
- ② Fundamental data: Last Observation Carried Forward (LOCF) from prior report period
- ③ If missing > 20 consecutive days: set NaN; exclude from IC calculation for that period

**STEP 7 CORPORATE ACTION ADJUSTMENT**

- ① Apply split & dividend adjustment factors from J-Quants or yfinance adjusted close
- ② All price series on total-return-equivalent basis — critical for momentum factors

**STEP 8 NORMALISATION (CS Z-SCORE → OPTIONAL TS Z-SCORE → CLIP $\pm 7\sigma$)**

- ① Cross-sectional z-score per date: $z_{f,i,t} = (f_{f,i,t} - \text{mean}_t) / \text{std}_t$
- ② Optional: time-series z-score per stock (stabilises long-run factor mean shifts)
- ③ Hard clip to $\pm 7\sigma$ — removes any remaining extreme outliers post-normalisation

**OUTPUT CLEAN FEATURE PANEL OUTPUT**

- ~480,000 observations: Daily × ~305 stocks × ~270 normalised features
- Each row carries: w_bench_{f,i,t} (from JPY121 ETF) and in_index_{f,i,t} ∈ {0, 1}
- Factor groups: Momentum · Volatility · Liquidity · Fundamental · Macro · Sentiment · Calendar

■ PHASE 1 — FEATURE SELECTION

STEP 9 UNIVARIATE IC RANKING → LOCKED 50-FEATURE SET

- ① Rank every feature by standalone cross-sectional Spearman IC vs forward returns
- ② Use purged walk-forward panel — no overlapping return windows in IC estimation
- ③ Keep top $K = 50$ features with highest mean $|IC|$ across history
- ④ Stability filter: IC sign must be consistent $\geq 70\%$ of months (discard noisy features)
- ⑤ Output: ONE locked 50-feature set — identical input to ALL downstream models



■ PHASE 2 — SIGNAL MODELS

STEP 10 TWO COMPLEMENTARY GRADIENT-BOOSTED MODELS

MODEL A — LGBMRanker

Loss: LambdaRank (pairwise ranking)

Learns relative ORDER of stocks

Target: which names beat the index

Covered IC ≈ 0.025 (in blend)

Robust to return-scale noise

MODEL B — LightGBM-Huber

Loss: Huber (L2 small + L1 large)

Predicts return SIZE, outlier-robust

Target: forward return magnitude

Covered IC ≈ 0.045 Uncovered IC ≈ 0.12 

STEP 11 SIGNAL ROUTING

- Covered stocks (full index members): Blend of Huber + Ranker (e.g. 60% / 40%)
- Uncovered stocks (extended universe, partial data): Huber only (more robust)



STEP 12 TRAINING PROTOCOL

- ① Walk-forward expanding window — training $t_{\text{start}} \rightarrow t_{\text{end}}$; purge gap; test $t_{\text{end}} + \text{gap}$ onward
- ② Covered model training: 2014-01 $\rightarrow$ 2017-06 (expanding); Uncovered: 2014-01 $\rightarrow$ 2018-06
- ③ Retrain monthly / quarterly — no future data ever enters training features



STEP 13 ALPHA CONSTRUCTION — GRINOLD-KAHN

- Grinold-Kahn: $\alpha_i = IC \times \sigma_i \times \text{score}_i$
- IC = realised signal IC · σ_i = annualised vol of stock i · score_i = z-scored ML output
- Converts dimensionless ML score $\rightarrow$ expected annual return in portfolio-optimiser units

■ PHASE 3 — PORTFOLIO OPTIMISATION (QP · CLARABEL)

STEP 14 QP OBJECTIVE FUNCTION

- Objective: maximise $\alpha^T \delta w - \lambda \cdot \delta w^T \Omega \delta w$
- α = Grinold-Kahn alpha vector · Ω = covariance matrix of active returns · λ = risk aversion

STEP 15 CONSTRAINTS C1 – C7

Portfolio Constraints	Risk & Cost Constraints
C1: $\Sigma w = 1$ (fully invested)	C3: β caps by cap tier (large / mid / small)
C2: $w_{\text{bench}} + \delta w \geq 0$ (long-only, no short single stocks)	C5: $TE \leq \text{budget}$ (annualised tracking error)
C4: $\delta w = 0$ if no signal (passive fallback)	C6: $ \text{sector } \Delta w \leq \text{cap}$ (sector exposure limit)
	C7: $\text{Turnover} \leq \text{budget}$ (transaction cost control)

STEP 16 ACTIVE WEIGHT OUTPUT

- Active weight: $\delta w_i = w_i - w_{\text{bench}_i}$ (deviation from JPY121 ETF benchmark weight)
- Constraint: $\Sigma \delta w_i = 0$ (active-neutral; overweights cancel underweights exactly)
- TE definition: $\sqrt{(\delta w^T \Omega \delta w)} \times \sqrt{252}$ | Target: $TE > 2\%$ annualised (5-day)

■ MARKET-NEUTRAL ACTIVE BOOK (NET = 0%)

STEP 17 LONG + SHORT CONSTRUCTION

■ LONG LEG	■ SHORT LEG	Active P&L
~225 NKY 225 constituents	NKY 225 index ~100%	$P\&L = \Sigma \delta w_i \cdot (r_i - r_{\text{index}})$
Fully invested, long-only	via futures / ETF (1321.T)	Purely from stock selection
Overweight high-alpha stocks	Hedges all market beta	Delta-neutral to NKY 225
$\delta w_i > 0 \rightarrow$ above benchmark	NET market exposure = 0%	No directional index exposure

OUTPUT LIVE PORTFOLIO — TARGET PERFORMANCE

- Rebalance: monthly · Live window: Jun 2018 → present
- Target CAGR: ~11.3% · Sharpe: ~5.76 · Max Drawdown: < 15%
- IC target: > 0.03 · ICIR > 0.5 · Tracking Error: 2–5% annualised
- Turnover: < 20% one-way per month · Sector Δw within cap